IT·STRESS RELIEF

SRE DA
LONDON 20

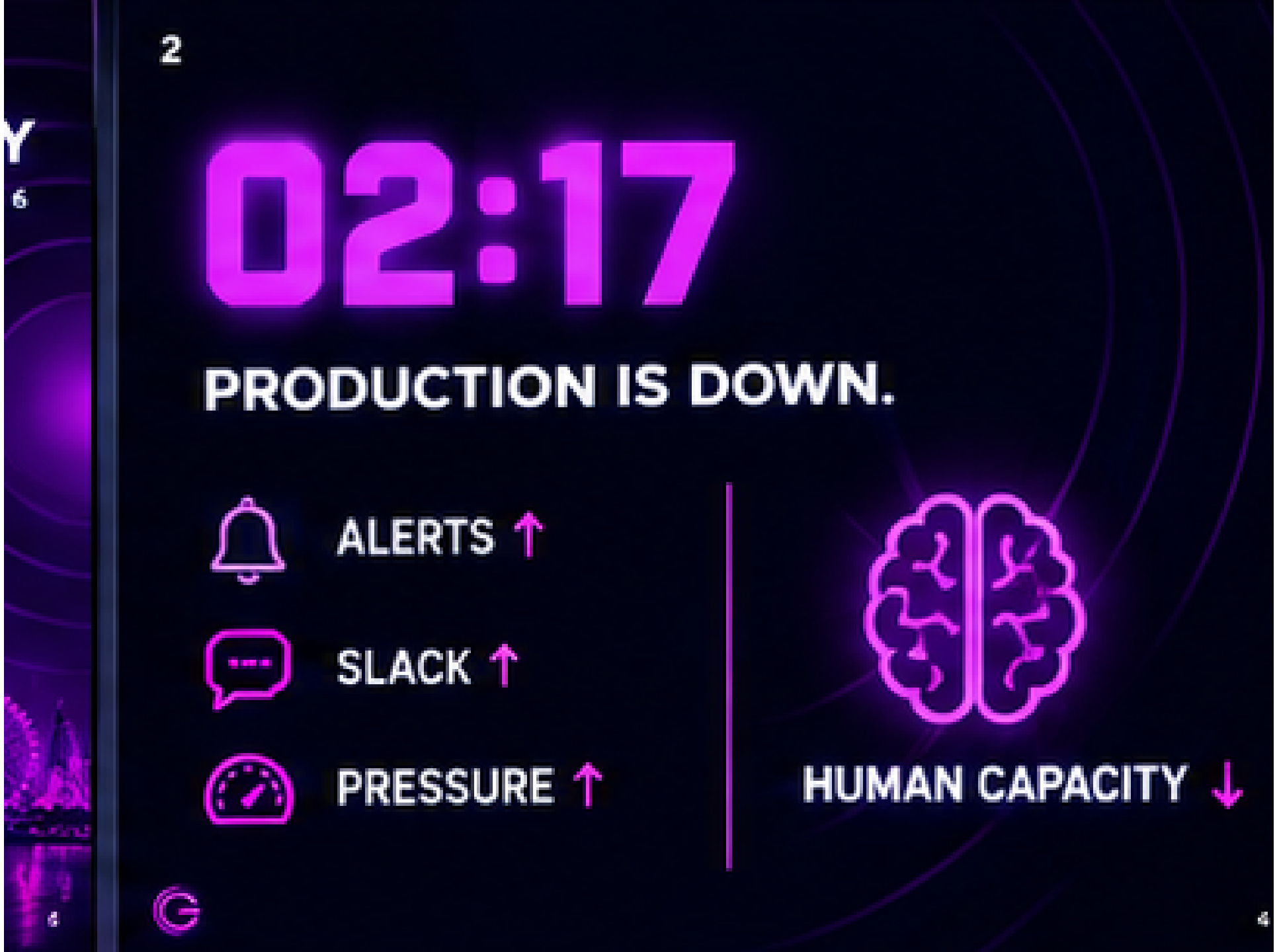
HUMAN SRE

Applying Reliability Engineering to the System Behind the Systems

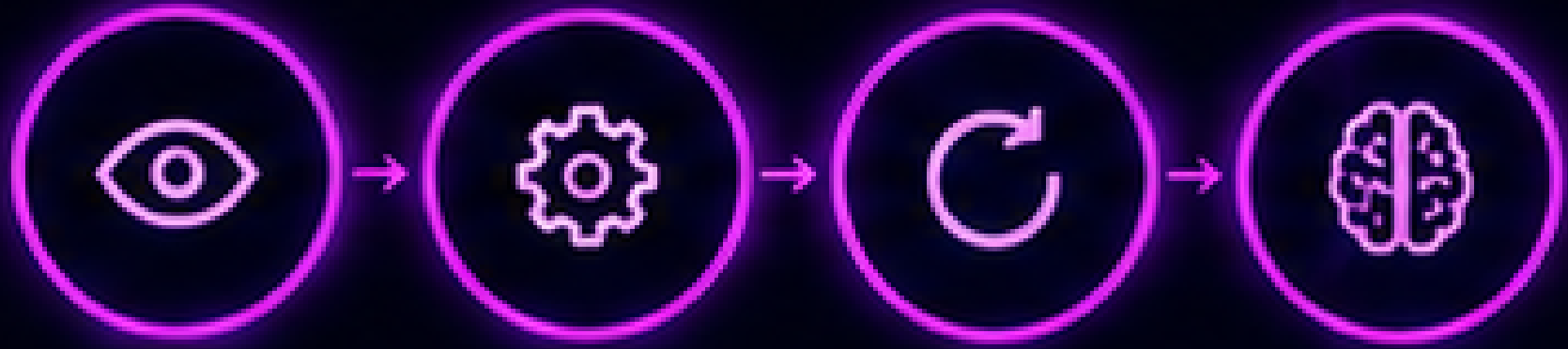
Gena Frangina
IT Stress Relief



SREday London · 24 September 2026


02:17
PRODUCTION IS DOWN.
ALERTS ↑
SLACK ↑
PRESSURE ↑
HUMAN CAPACITY ↓

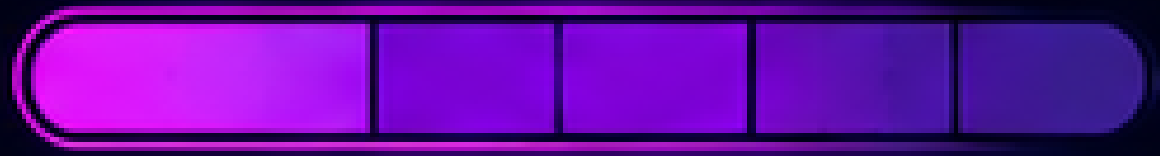
WE ALREADY KNOW HOW TO DESIGN FOR FAILURE.



What if the same thinking applied to humans?

HUMANS HAVE CAPACITY LIMITS

CAPACITY ≠ ∞



HIGH LOAD



INTERRUPTIONS



CONTEXT SWITCHING



UNCERTAINTY



FATIGUE



DECISION LOAD

LOAD ↑ → PERFORMANCE DEGRADES

WHAT DOES YOUR TELEMETRY LOOK LIKE?

SYSTEM		HUMAN
Latency ↑	→	Thinking slows
Error rate ↑	→	Mistakes increase
Saturation ↑	→	Cognitive overload
Alert volume ↑	→	Constant interruption
Resource exhaustion	→	Fatigue
Degraded service	→	Poorer decisions



WHAT'S YOUR FIRST WARNING SIGNAL?

A SYSTEM CAN BE DEGRADED BEFORE IT FAILS.

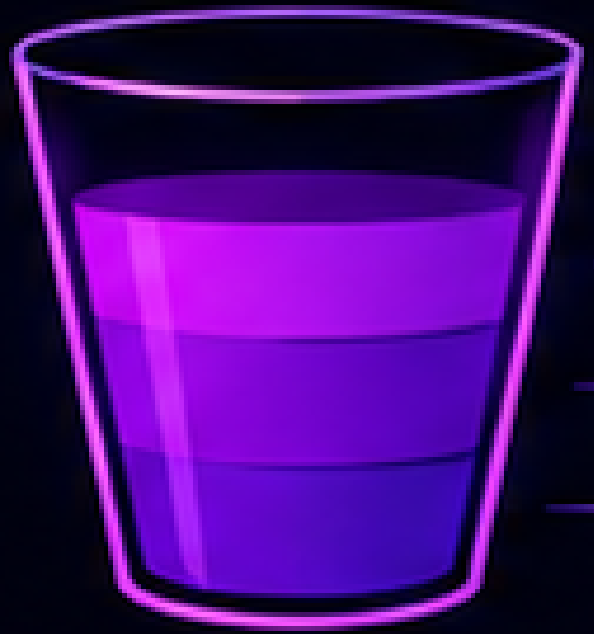


FUNCTIONING ≠ RELIABLE

The human system is no different.

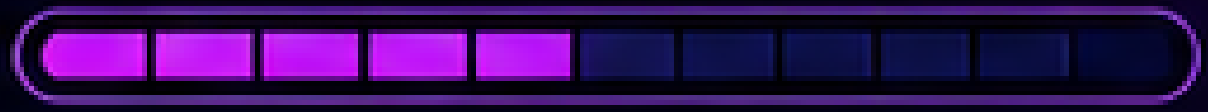
YOUR CAPACITY IS FINITE.

Where should you spend it?

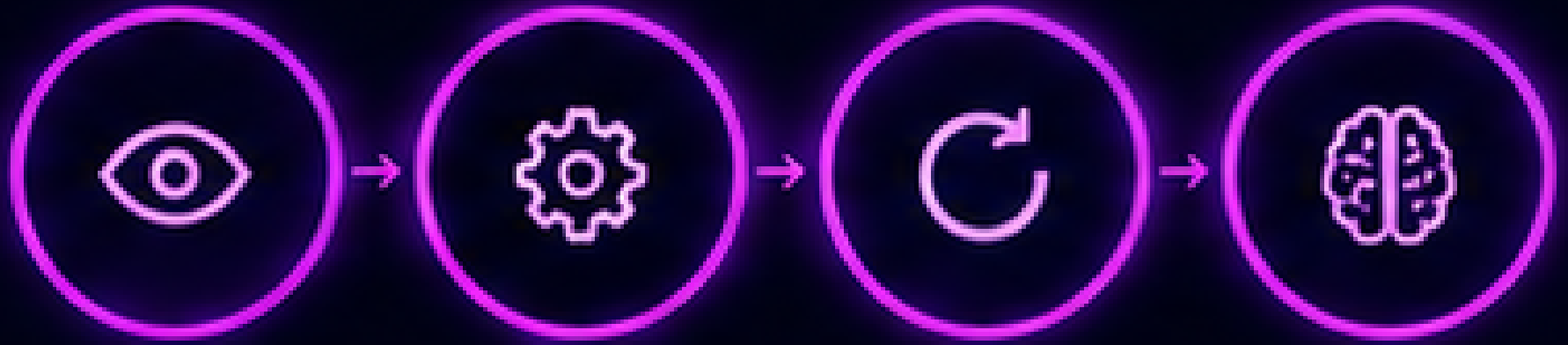


- INCIDENTS
- INTERRUPTIONS
- DECISIONS
- CONTEXT SWITCHING

CAPACITY



HUMAN SRE



OBSERVE

Notice the signals

RESPOND

Reduce unnecessary load

RECOVER

Return towards baseline

LEARN

Improve the system

RECOVERY IS PART OF RELIABILITY.



Not:

INCIDENT → RESPONSE → MORE WORK → MORE WORK → MORE WORK

AFTER THE INCIDENT:

~~WHY DIDN'T I COPE BETTER?~~

WHAT CONDITIONS MADE THIS MORE LIKELY?

- WORKLOAD
- ON-CALL DESIGN
- INTERRUPTIONS
- AMBIGUITY
- STAFFING
- COMMUNICATION
- RECOVERY

Add system responsibility.

11

WHERE DOES AI FIT?

COGNITIVE SUPPORT

BEFORE

- Structure information
- Identify unknowns

DURING / AFTER

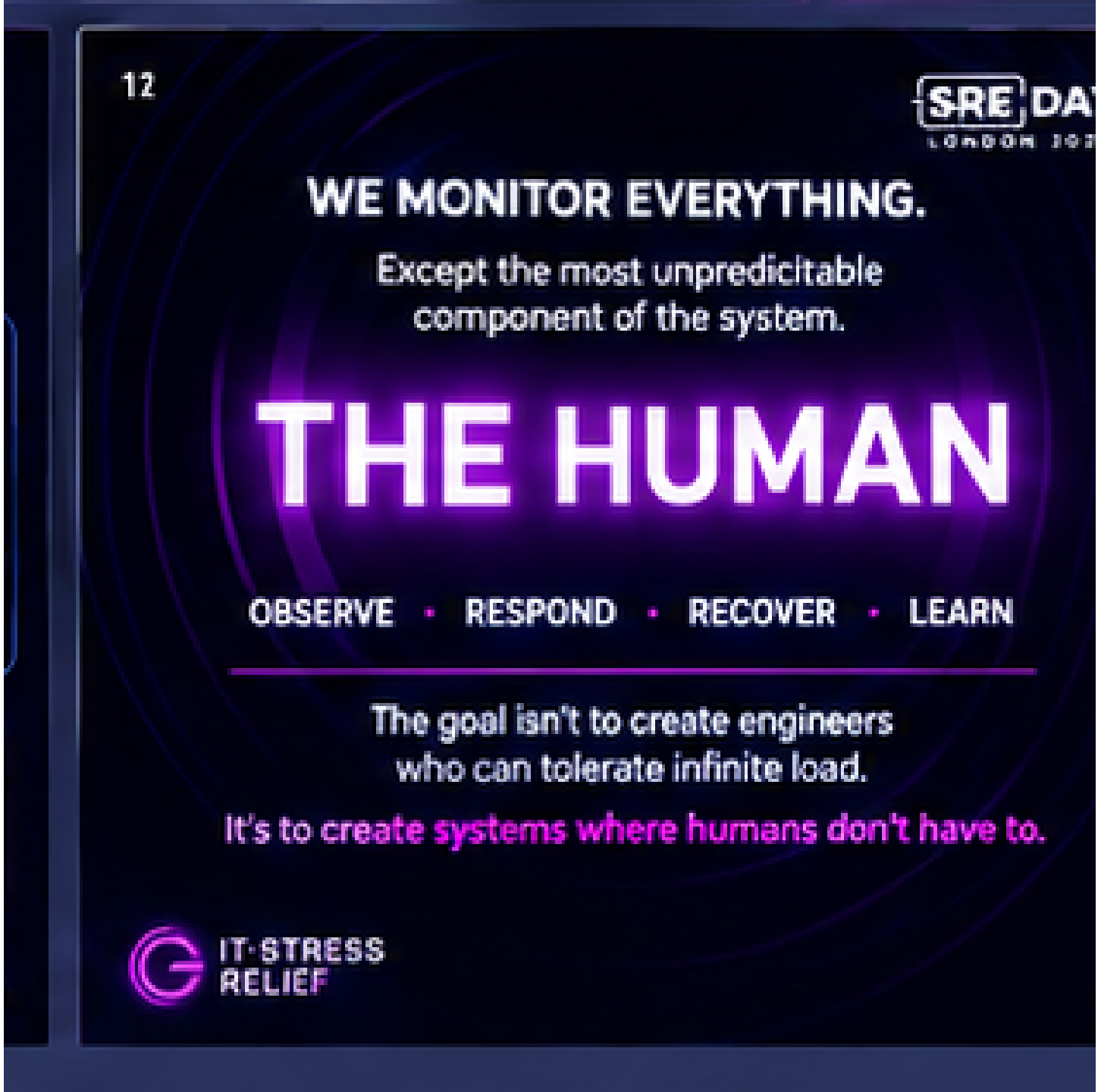
- Summarise
- Separate facts from assumptions

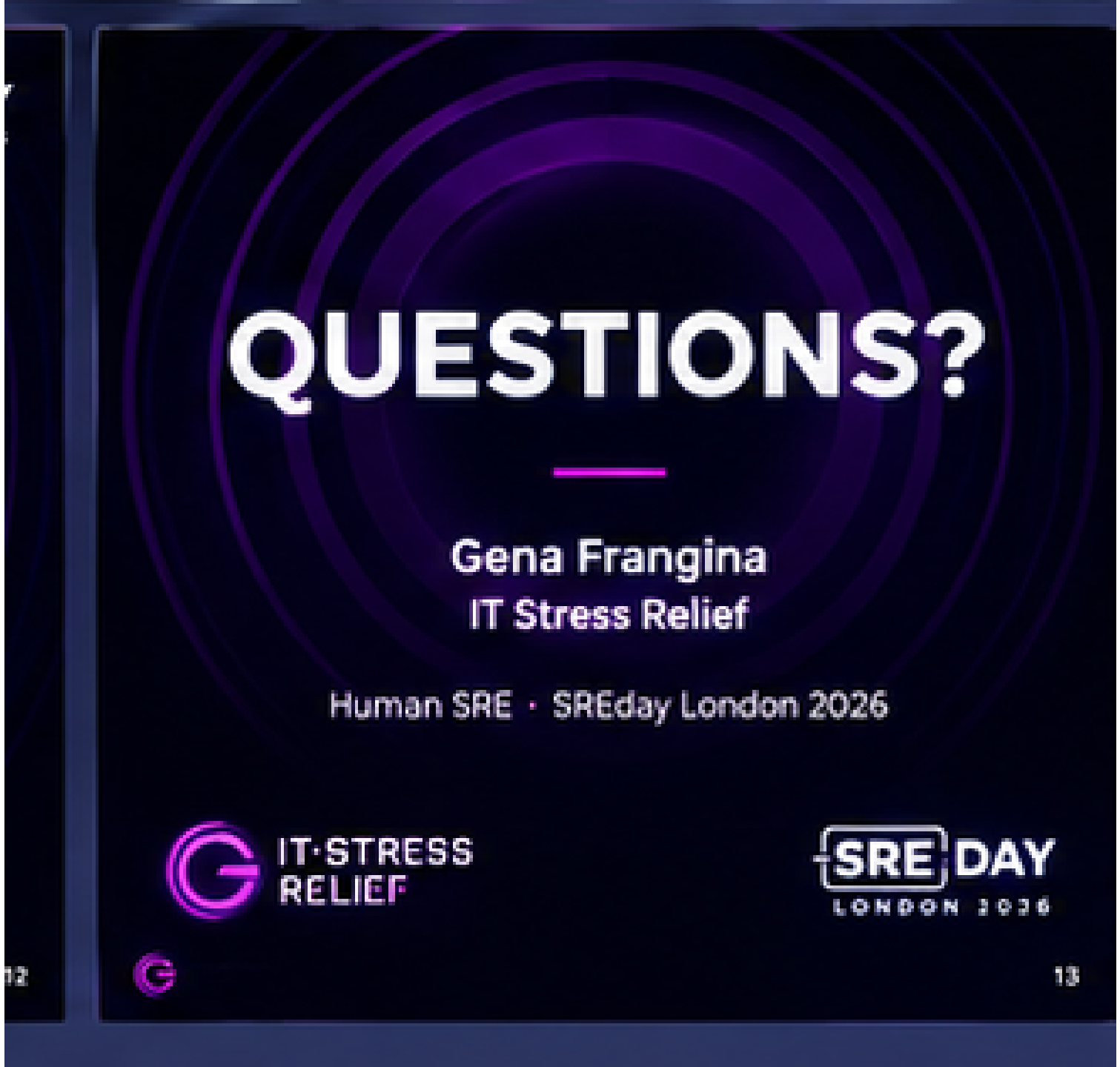
RECOVERY

- Reflect
- Create a transition shutdown ritual

REDUCE COGNITIVE LOAD.

DON'T NORMALISE UNHEALTHY LOAD.


12
SRE DA
LONDON
WE MONITOR EVERYTHING.
Except the most unpredictable
component of the system.
THE HUMAN
OBSERVE · RESPOND · RECOVER · LEARN
The goal isn't to create engineers
who can tolerate infinite load.
It's to create systems where humans don't have to.
IT·STRESS
RELIEF


QUESTIONS?
Gena Frangina
IT Stress Relief
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13